



Bending backbone

Running from your skull down to your hips is a long chain of bones, forming your backbone. This strong column of bones holds your head and body upright but at the same time allows your neck and back to twist and bend. Your backbone also protects your delicate spinal cord—a large bundle of nerve fibers that sends messages to and from your brain to the rest of your body.

STACK OF BONES

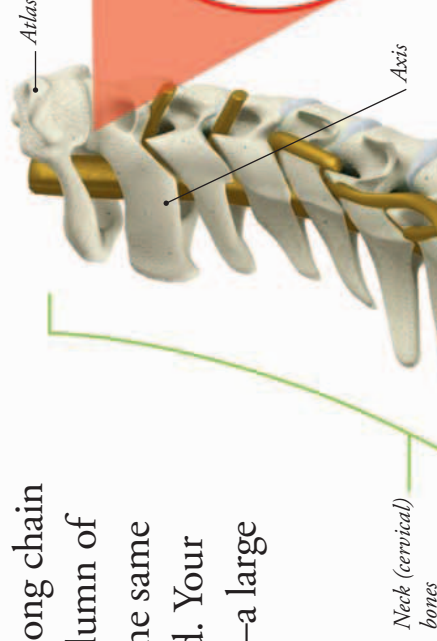
Twenty-six ring-shaped bones called vertebrae form your S-shaped backbone. There are seven neck bones supporting your head and 12 chest bones that are linked to your ribs. Your lower back has five lumbar bones that bear most of the weight of your upper body, and a group of small bones in the sacrum and coccyx are fused together to form the tail-end of your backbone.

Wow!

You have the same number of neck bones as a giraffe—the difference being that a giraffe's neck bones are much longer.

▼ ATLAS AND AXIS

The atlas, the topmost bone, allows your head to nod. It also rotates on the axis bone beneath it so your head can turn from side to side.



FLEXIBLE SPINE

The joints between any two vertebrae allow only limited movements. But added together, these small movements make your backbone very flexible. It can bend forward, backward, from side to side, and twist and turn. The backbone's soft curve gives it the springiness to absorb jolts as you move.

A hole through each bone forms a tunnel-like area for the spinal cord.

Nerve roots exit through narrow spaces along the backbone.

Your backbone curves gently, protecting the spinal cord from snapping.

